

GOPIKA P

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CAREER OBJECTIVE

Motivated **Hardware Engineer** skilled in PCB design and layout, electronics components, circuit design, microcontrollers, Embedded C, Digital Electronics, Verilog and SystemVerilog. Seeking internship and job opportunities primarily in the PCB Design, Embedded Systems domain and RTL Design Engineer where I can apply my hardware and programming skills to develop reliable, efficient and real-world electronic solutions.

EDUCATION

Qualification	Institution	Year	Score
B.E. Electronics and Communication Engineering	Kongunadu College of Engineering and Technology, Thottiyam	2023-Present	94%
HSC	Trichy Government Model Higher Secondary School, Trichy	2023	70%
SSLC	Government Higher Secondary School, Pappapatti	2021	All Pass

INTERNSHIP

1. **Embedded Systems - CodeBind Technologies, Trichy** **15 days**
   Worked on sensors, actuators and also learned about Arduino IDE and done a project in the title of "Disaster Alerting System" using the sensors.
2. **Embedded Systems & IoT – Tara's Systems and Solutions, Bengaluru** **6 Months**
   Worked on MSP430 and STM32 Microcontroller and build a project on "Smart Priority-based Load Control System" using STM32.
3. **VLSI - Univision, Chennai** **1 Month**
   Learned about Verilog HDL and SystemVerilog & UVM basics

TECHNICAL SKILLS

- **PCB Design:** Semiconductor Components, KiCad, Schematic Capture, PCB Layout, Component Placement, Routing, Footprint Creation, DRC.
- **Embedded:** Hands-on experience in Arduino, ESP32, STM32, MSP430 & FPGA board.
- **Electronics:** Digital Electronics, Circuit Design, Electronic Components, SMD & Through-Hole Components.
- **Programming:** C, Verilog & SystemVerilog (Synthesizable concepts).

SOFT SKILLS

- Technical Curiosity
- Learning Agility
- Good Communication

PROJECTS

Project 1

Title: *IoT-Based Urine Bag Monitoring System*

Tools Used: Load Cell + HX711 + ESP32 to monitor urine level in real time

Description: A smart healthcare solution using a load cell and ESP32 to continuously monitor urine volume and automatically alert nurses via mobile app, LED, and buzzer when the bag reaches a critical level.

Project 2

Title: *Smart Priority-based Load Control System Using STM32*

Tools Used: STM32CubeIDE, STM32WB55 Microcontroller, ACS712 Current Sensor, Relay Module, Load (LED / Fan) and Buzzer

Description: Intelligently monitors electrical load and based on priority it automatically disconnects low-priority loads during overload while maintaining critical loads for safe and efficient power management.

CERTIFICATIONS

- ✓ PCB - Value Added Course and Designed a Power Supply PCB board and viewed in 3D.
- ✓ PLC - Value Added Course
- ✓ NPTEL Certification – Introduction to Internet of Things (2025) with 5% topper ranking (90 marks)
- ✓ TYPE WRITING - Secured First Class with Distinction in English Junior Typewriting.

HACKATHONS, ACHIEVEMENTS & PRESENTATIONS

- Received a **Cash Award for securing class 2nd** in academic subjects upto 5th semester and **for securing class 1st** in 6th semester.
- Secured **First Prize with Cash Award** in Project Presentation for the project **"Smart IoT Based Urine Bag Monitoring System"** at Kongu Engineering College and KPR College.
- Secured **Second Prize with Cash Award** in Idea Presentation.
- Participated in the **Hackathons** like SIH, MSME, Teliport Season 3 organized by Tata Elxsi.

AREA OF INTERESTS

- ❖ PCB
- ❖ Embedded Systems
- ❖ Internet of Things (IoT)
- ❖ VLSI (RTL Digital Design)

TOOLS & SOFTWARE

- ❖ OrCad & KiCad
- ❖ STM32Cube IDE
- ❖ Proteus
- ❖ ModelSim & Vivado

PROFESSIONAL MEMBERSHIPS

- Active Student Member of Indian Society for Technical Education (ISTE).
- Student Member of Institution of Electronics and Telecommunication Engineers (IETE).

HOBBIES

- ➢ Singing & Gardening
- ➢ Exploring new technologies

DECLARATION

I hereby declare that the information provided above is true to the best of my knowledge.

Place:

Date:

SIGNATURE

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